

Keeper Consistency Audit — Three-Generations Mechanism Reconciliation

Keeper

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(K-audit number to be assigned against the authoritative chain at EOD. The K194-K217 range is occupied; this filing does not claim a specific number to avoid collision — the finding stands regardless of bookkeeping.)

What’s strong (acknowledge first)

- The two-region framework (Lyra_Bulk_Vs_Shilov_Formal_Investigation_v0_1.md) is sound at FRAMEWORK tier: Shilov $S^4 \times S^1$ vs bulk interior is rigorous Cartan structure; the lepton↔Shilov / quark↔bulk *assignment* is an honest Casey-directed input; the Ogg-vs-Mersenne arithmetic split is empirically grounded and correctly flagged as substrate-mechanism CANDIDATE (Section 2). Lyra’s Section 7 honest-scope is well-disciplined.
- The quark-confinement-via-bulk→Shilov-projection candidate (Section 3.3) is a genuinely interesting INTERPRETIVE mechanism. No overclaim — flagged as candidate.

This audit does not dispute any of that. It isolates one consistency defect.

The finding (MODERATE — gap in argument, not a broken proof)

The Wednesday corpus contains three distinct, mutually unreconciled mechanisms for “why exactly three generations,” and they are not the same claim. At least one was already flagged by Casey as numerological and asked to be dropped. None can go into the paper series until reconciled.

#	Mechanism	Source doc	“3” comes from	Status
A	N _c =3 color directions (R/G/B) projected out → e/μ/τ; all at the SAME Wallach K-type V _(1/2,1/2)	Lyra_Lepton_K_Type_Identification.md Sec 2.2	Ident. Bification	Casey flagged “N _c =3 → 3 generations” as numerological coincidence Wednesday (RUNNING_NOTES Wed item 7); asked Lyra to drop it. This doc still carries it.

#	Mechanism	Source doc	“3” comes from	Status
B	Cal #139 cyclotomic chain has 4 elements {rank, N_c, n_C, g}; minus rank as base/seed = 3; $X \in$ {N_c, n_C, g} $\rightarrow$ 3 generations	Three-lepton- generations refined track (Casey-endorsed)	4 chain elements – 1 base	Casey-endorsed refined version; 2 SVC gates open
C	One substrate K-type per generation manifests on BOTH regions via Hardy space $H^2(D_{IV}^5)$: lepton = boundary value, quark = bulk interior of the SAME K-type	Wednesday evening synthesis (arc v0.2)	(addresses lepton $\leftrightarrow$ quark UNITY, not the count directly)	INTERPRETIVE; strongest + riskiest

Why this is a defect, not just parallel views

1. A and B give different count-mechanisms. A ties the count to $N_c=3$ (color). B ties it to the chain length ($4-1=3$). These are different substrate causes that happen to both yield 3. Casey already rejected A’s form. A paper cannot present both as “the” mechanism.
2. A contradicts B’s own structure. A puts all three generations at the *same* Wallach K-type $V_{(1/2,1/2)}$, distinguished only by a color-projection label. B distinguishes generations by *different chain exponents* $X \in \{N_c, n_C, g\}$ — i.e., genuinely different substrate data per generation. These are incompatible pictures of what a generation *is*.
3. C, if taken literally, has a heavy consequence that neither A nor B addresses. If lepton and quark of a generation are boundary value and bulk interior of ONE holomorphic K-type, then Hardy-space determinacy makes them *not independent* — the boundary value fixes the interior. That implies a rigid within-generation lepton $\leftrightarrow$ quark mass linkage. That is a strong, falsifiable claim (is m_e tied to m_u/m_d by a boundary-value relation?), and it is currently asserted as “the unification mechanism” without that consequence being derived or tested. C is also in tension with A (A says leptons are atomic Shilov K-types with NO bulk/internal structure; C says the same K-type HAS a bulk interior = the quark).

Severity: MODERATE

Not CRITICAL — no proof is broken; the two-region framework stands. But it is more than MINOR: three competing generation mechanisms, one already-rejected, heading into a 14-paper series is a consistency hazard that an external referee will catch immediately and that would damage credibility. It must be resolved before Wave 1 papers (esp. B1 WCGP, B4 Lepton 3-Gen) reach v0.2.

Recommended reconciliation (for Lyra, gated by Cal cold-read)

1. Retire mechanism A ($N_c=3$ color-projection $\rightarrow$ 3 generations) explicitly, per Casey’s Wednesday call. Update `Lyra_Lepton_K_Type_Identification.md` to remove or demote the color-direction-count framing to “superseded; see Cal #139 chain.” Keep the color-singlet *projection* content (that leptons are color-singlet is fine); drop only the “ N_c counts the generations” inference.
2. Adopt mechanism B (Cal #139 chain, $4-1=3$) as the canonical generation-count mechanism. It is Casey-endorsed and has genuine per-generation substrate data (different X).

3. Re-scope mechanism C as a claim about lepton $\leftrightarrow$ quark *unity within a generation*, NOT about the count. Then state its heavy consequence explicitly and make it falsifiable: does Hardy-space determinacy predict a within-generation lepton $\leftrightarrow$ quark mass relation? If yes $\rightarrow$ strong prediction (test it). If the relation is too rigid to match data $\rightarrow$ C must weaken from “same K-type” to “paired K-types,” which is the safer afternoon framing. Do not let C enter a paper as “the unification mechanism” until this is settled.
4. Confirm B and C compose: the canonical picture should be “B sets the count (3 chain levels); C (if it survives step 3) sets the lepton $\leftrightarrow$ quark relationship per level.” A is gone.

Disposition

- Two-region framework: CONDITIONAL PASS, FRAMEWORK tier. Proceed.
- “Three generations” as a settled result: FAIL pending reconciliation. Hold at OPEN-PROBLEM until steps 1-4 close.
- “One-K-type-per-generation” as a Casey-named principle (queued Thursday decision): recommend NOT YET. It is mechanism C, which is not yet reconciled with B and carries an untested heavy consequence. Naming it now would lock in an unverified claim. Revisit after step 3.
- Feeds Casey decision queue: this is the substance behind “one-K-type-per-generation as Casey-named principle?” — Keeper answer is hold.

Honest scope

This audit asserts a *consistency* problem across three Wednesday docs; it does not claim any of the three mechanisms is individually wrong. B is the most defensible. C may be a real and beautiful unification OR an overreach of Hardy-space determinacy — that is exactly what step 3 tests. The point of the audit is that we cannot carry all three unreconciled, and we must not name C as a principle before it is tested. Quaker discipline: the most elegant claim (C, “lepton and quark are one object”) gets the most scrutiny, not the most enthusiasm.

— Keeper, 2026-05-28 Thursday (v0.1). K-number pending registry assignment.

v0.2 Addendum — Morning convergence (Cal #146 + Lyra Coxeter finding)

Within hours of filing v0.1, two independent results landed that resolve most of this audit. I record them and update every disposition. Two of them correct my own framing — I own that below.

Correction I accept (Cal #146)

Cal cold-read the unification framework and caught that my Thursday morning broadcast overstated it. I wrote “one K-type per generation manifesting as both a lepton and a quark.” Per Lyra’s own taxonomy (Lyra_Comprehensive_Particle_Taxonomy_v0_1.md §2.1): - e_L = (Shilov, odd, L, -1, W₀) - u_L = (Bulk, odd, L, +2/3, W₀)

Gen-1 lepton and gen-1 quark are distinct K-types that share only the winding-mode coordinate W₀ (= the generation index). They differ in Region AND Charge. So mechanism C is NOT “one K-type, two manifestations” — it is “distinct K-types sharing a generation/winding-mode axis.”

This is the correct framing and I was wrong in the broadcast. It is also *better*: it dissolves exactly the heavy consequence I flagged in v0.1 (the rigid within-generation lepton $\leftrightarrow$ quark mass linkage). Distinct K-types $\rightarrow$ distinct Casimir $\rightarrow$ distinct masses, naturally. The mass problem I worried about was an artifact of my overstatement, not of the underlying structure. Cal’s finding is correct; I’m adopting it.

Strengthening that landed (Lyra Coxeter-number finding)

Lyra extended to the affine quantum group $U_q^{+(B_2)}(1)$ and found mechanism B's count is not numerological at all — it is root-system geometry: - Cal #139 chain length = Coxeter number $h(B_2) = 4 - 3$ generations = $h(B_2) - 1$ (chain minus base/seed) - 3 colors = dual Coxeter number $h^\vee(B_2) = 3 = N_c$

Generation count comes from h ; color count comes from $h^\vee$. Different invariants. This is the principled grounding mechanism B lacked, and it explains the chain length itself (previously an open question per Cal #139). Honest scope per Lyra: Serre-relation level rigorous; full Phase 0 closure still needs PBW + Macdonald structure constants + $A_{\text{sub}} \leftrightarrow U_q^{+(B_2)}$ isomorphism. FRAMEWORK-PLUS, cross-checks the existing over-determination.

Updated dispositions

Mechanism	v0.1 disposition	v0.2 disposition
A (N_c color-projection $\rightarrow$ generations)	retire (numerological, Casey-flagged)	RETIRE — now positively superseded, not just rejected. Coxeter grounding shows generations come from $h(B_2)$, colors from $h^\vee(B_2)$: two different invariants. A conflated them.
B (chain $4-1=3$)	adopt as canonical	ADOPT + STRENGTHENED — chain length = $h(B_2)=4$ is now <i>explained</i> , not assumed. Generations = $h-1$, colors = $h^\vee$.
C (one K-type dual manifestation)	INTERPRETIVE, risky, untested heavy consequence	REFRAMED to “distinct K-types sharing winding-mode/generation axis W_0 ” (Cal #146). Heavy consequence dissolves. Now consistent with B (W_0 = the chain/generation index).

The three mechanisms now reconcile into one clean picture: B_2 root-system geometry sets the counts ($h \rightarrow$ generations, $h^\vee \rightarrow$ colors); a generation is one value of the winding-mode axis W_0 shared across distinct K-types that differ by region/charge; leptons and quarks of a generation are *related by sharing W_0* , not identical. A is gone. C is corrected. B is grounded.

Updated recommendation on the Casey-named-principle decision

v0.1 said HOLD on “one-K-type-per-generation.” Revised: the *corrected* statement — “Generation = the shared winding-mode axis (W_0) of the B_2 affine structure; counts forced by Coxeter $h(B_2)=4$ and $h^\vee(B_2)=3$ ” — is now clean, non-numerological, and consistent. It is more nameable than what I cautioned against. But still gate naming on: (i) chain-termination / PBW closure ($h(B_2)=4$ must force chain termination, not just match it); (ii) Cal cold-read of the Coxeter finding; (iii) Elie numerical verification. Recommend Casey hold the name one more cycle until (i)-(iii) close, then it is a strong Casey-named candidate (“Coxeter Generation Principle” or similar). This is days, not weeks.

Calibration #30 — I endorse the 7th-instance promotion

Cal flagged: if I reframe “one K-type” $\rightarrow$ “shared winding mode, distinct K-types,” that is a 7th honest-negative-strengthens instance in a structurally-distinct context (particle taxonomy, not the cyclotomic chain), satisfying the Calibration #30 candidate STANDING promotion gate. I am making the reframe (above). So the 7th instance

is real: my overstatement was caught, the correction made the framework *cleaner and more correct*, and it came from a new domain. Keeper endorses promoting Calibration #30 to STANDING. Cal's call on final wording.

— Keeper, 2026-05-28 Thursday (v0.2). Audit substantially resolved by morning convergence; residual gate is chain-termination-from-Coxeter + Cal cold-read + Elie numerics.